



COURSE SLIDES

# Google Workspace with Gemini

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Course Code: C155

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

# Welcome & Housekeeping

# Course Trainer Profiles

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1

## Truman Ng

ACTA-certified trainer and technology practitioner with extensive experience in cloud, networking, databases, web systems and enterprise implementation.

# Let's Know Each Other

- 1 Your name, role and the Workspace app you use most often.
- 2 One repetitive writing, analysis or coordination task you would like to improve.
- 3 One data, accuracy or sharing risk that should never be automated away.

# Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

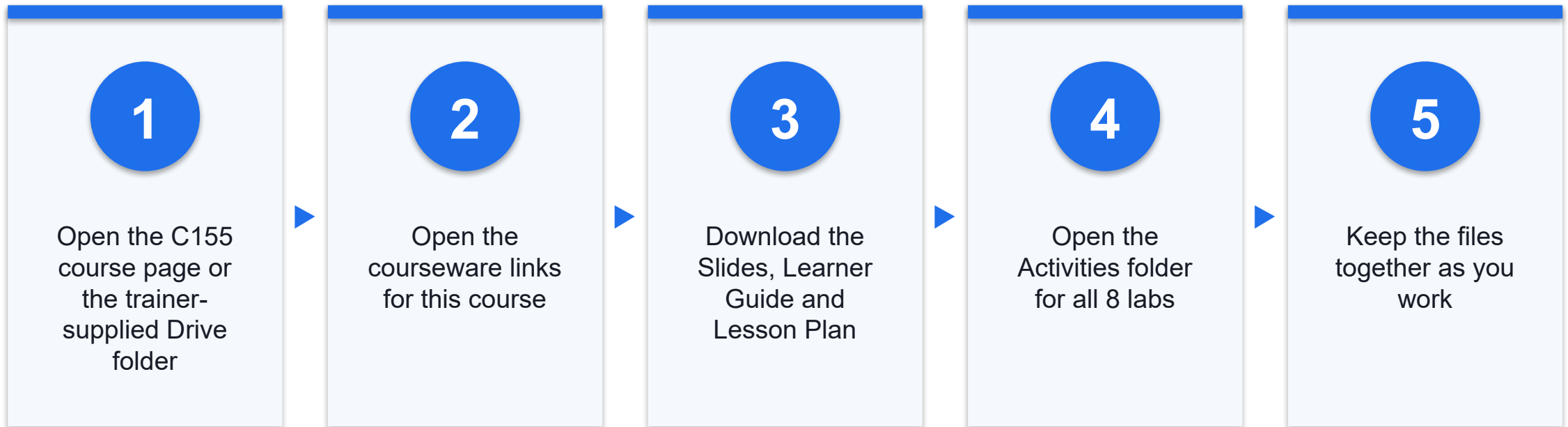
Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

# Access Course Materials

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**Course page:** <https://www.tertiarycourses.com.sg/google-workspace-with-gemini.html>

# Lesson Plan — 2 Days, 15 instructional hours

### Course sequence

- Day 1 — Prompt safely, then create and refine a partner communication pack
  - Topic 1: Getting Started with Gemini in Google Workspace (Labs 1–2)
  - Topic 2: Gemini in Docs, Slides and Gmail (Labs 3–4)

### Day 2

- Day 2 — Turn clean data and meeting evidence into a controlled cross-app workflow
  - Topic 3: Gemini in Sheets and Data Analysis (Labs 5–6)
  - Topic 4: AI-Powered Workflows and Automation in Workspace (Labs 7–8)
- Daily timing
  - 9:30 am - 6:30 pm (1-hour lunch; two 15-minute tea breaks; 7.5 instructional hours)

# Learning Outcomes

1

## Prompt & Protect

Explain how Gemini works across Google Workspace, write effective prompts and apply responsible data, privacy and verification controls.

2

## Write & Present

Draft, rewrite, summarise and refine workplace content in Docs, Slides and Gmail while preserving source accuracy, audience fit and human approval.

3

## Analyse & Visualise

Clean, organise, analyse and visualise spreadsheet data with Gemini in Sheets, formulas and reproducible checks.

4

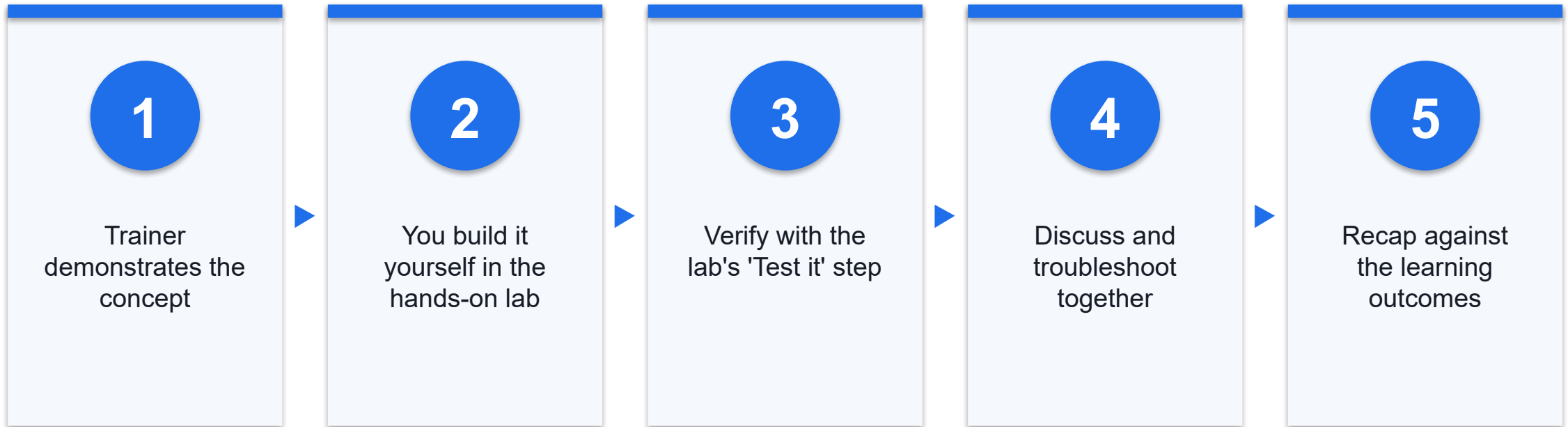
## Connect & Automate

Design an AI-assisted workflow across Meet, Drive, Sheets, Docs and Gmail, then implement and verify a bounded Workspace automation.



# How You'll Learn

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## CORE CONCEPTS

# Work Smarter Without Giving Up Control

# Four Ideas That Govern the Course

1

## Candidate, not conclusion

Gemini creates a first pass; a human still verifies and decides.

2

## Permission-aware context

Use the minimum approved Workspace sources and keep sharing deliberate.

3

## Visible evidence

Sources, formulas, logs and review notes make work reproducible.

4

## Bounded action

Draft, test and observe before any workflow scales or reaches others.

# The P-T-C-F + G-E-A-R Pattern

1

## PERSONA + TASK

Who needs help and what finished action or artifact is required?

2

## CONTEXT + FORMAT

Which approved sources, constraints and output structure govern the response?

3

## GROUNDING + EVIDENCE

Can every material claim, number and action be traced and reproduced?

4

## AUDIENCE + RISK

Is the result clear, permitted, appropriately shared and ready for human approval?

## COURSE PRINCIPLE

**AI speed is useful only when the evidence and decision owner stay visible.**

Across every Workspace app, the durable workflow is select approved context, generate a bounded candidate, verify it, and act through normal controls.

# What You Will Build

## Communication pack

- Prompt charter and verified briefing
- Docs, Gmail and Slides artifacts

## Data decision pack

- Clean registration tracker
- Metrics, pivot and chart

## Workflow pack

- Reviewed meeting recap
- Drive structure and controlled automation

# How Every Lab Progresses

1

Start from the previous checkpoint and the same synthetic Aster & Finch partner-briefing scenario.

2

Use a structured prompt or deterministic rule against named approved sources.

3

Create an observable artifact inside Workspace or a supplied fallback.

4

Run a Test It check, record exceptions and save the checkpoint for the next lab.

# Human Checks Across the Workspace Flow

1

## INPUT

Authority, data class, minimum fields, file permissions and source version.

2

## GENERATION

Prompt contract, source boundary, uncertainty and unsuitable-content check.

3

## TRANSFORMATION

Formula logic, row grain, duplicates, ranges and reproducible rules.

4

## OUTPUT

Fact review, recipients, sharing, accessibility, approvals and audit evidence.



TOPIC 01

# Getting Started with Gemini in Google Workspace

Generative AI and Gemini - Access across apps - Effective prompting - Responsible, secure and private use

## Key Concepts — Getting Started with Gemini in Google Workspace

1

### Generative AI

A probabilistic system that creates a candidate output from instructions and context; it does not guarantee truth.

2

### Workspace grounding

Gemini can use selected Workspace sources and the user's existing permissions to add relevant context.

3

### P-T-C-F prompt

Persona, Task, Context and Format make the request, evidence boundary and expected output explicit.

4

### Human review gate

A named person verifies facts, privacy, permissions, tone and destination before output is used.

# Introduction to Generative AI and Gemini in Workspace

## WHAT

- Generative AI predicts and assembles a useful candidate response from patterns in data, instructions and supplied context. Gemini in Google Workspace places this capability inside familiar apps so a user can draft, summarise, organise and analyse without treating the generated output as an authoritative record.

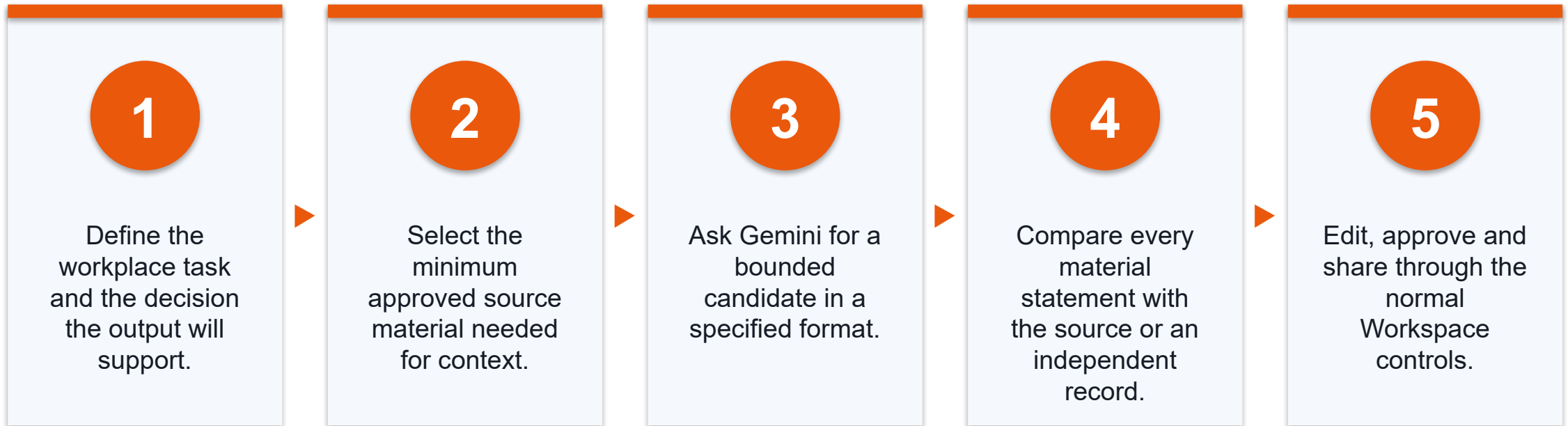
## WHY IT MATTERS

- The productivity gain comes from reducing the first-pass effort, not from transferring accountability to the model. A useful mental model is assistant, source and reviewer: Gemini proposes, approved Workspace material grounds the task, and a person decides what is accurate and appropriate.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Introduction to Generative AI and Gemini in Workspace



## In Practice — Introduction to Generative AI and Gemini in Workspace

### WORKED EXAMPLE

- Aster & Finch is planning a synthetic partner briefing and has an approved project brief.
- Gemini converts the brief into a draft agenda, but labels missing venue and speaker details as open questions.
- The coordinator verifies those fields before the agenda becomes a shared document.

### DECISION GUIDE

- **Use when**
  - A repetitive drafting or synthesis task has approved source material and a human owner.
  - The output can be checked before it influences a customer, employee or business decision.
- **Avoid when**
  - The task needs a guaranteed fact but no trustworthy source is available.
  - The user intends to send or publish the first response without review.

## Quality Lens — Introduction to Generative AI and Gemini in Workspace

### FAILURE SIGNAL

- The response sounds confident but cannot show where a key statement came from.

### REPAIR MOVE

- Constrain the prompt to named sources and require UNKNOWN for unsupported details.

### QUALITY EVIDENCE

- A reviewer can trace each material statement to an approved file or record.

# Setting Up and Accessing Gemini Across Workspace Apps

## WHAT

- Gemini features appear according to the user's Google account, eligible plan, administrator settings, language, region and app. Common entry points include an Ask Gemini side panel or an in-context create, refine, summarise or visualise control; exact labels may change as Workspace evolves.

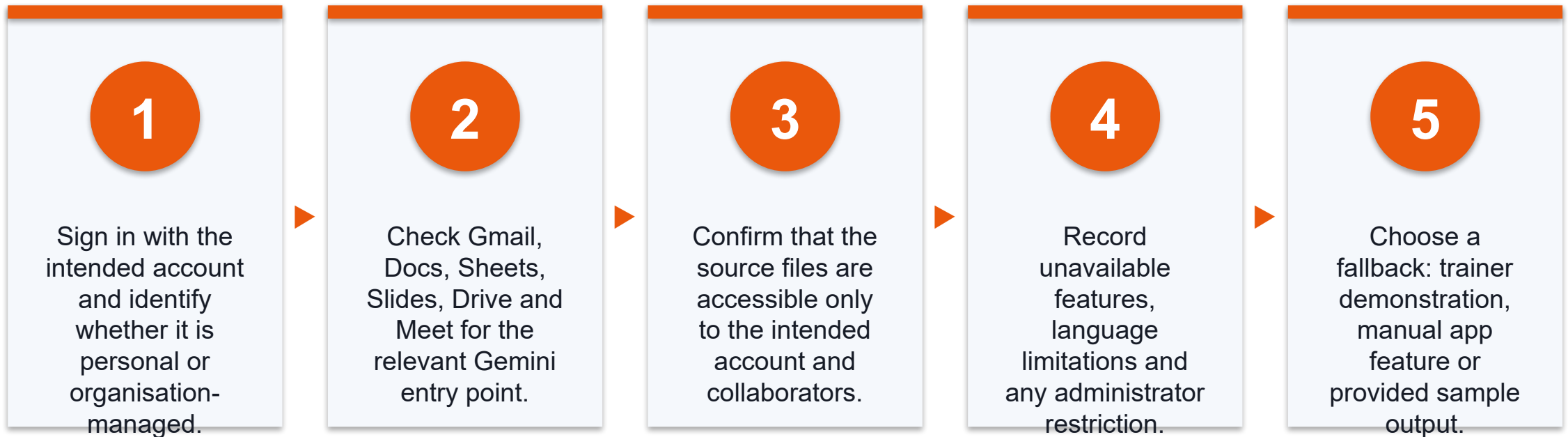
## WHY IT MATTERS

- A workflow that assumes every account has the same controls will fail in a real class or organisation. Separating the durable job from a particular button lets users verify availability, protect permissions and use a manual or trainer-provided fallback without losing the learning objective.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Setting Up and Accessing Gemini Across Workspace Apps





## In Practice — Setting Up and Accessing Gemini Across Workspace Apps

### WORKED EXAMPLE

- A learner can open Gemini in Docs and Sheets but not Take notes for me in Meet.
- The learner records the gap, uses the supplied meeting transcript and still completes the notes-review workflow.
- No account credentials or protected organisation files are moved to another service.

### DECISION GUIDE

- **Use when**
  - You are validating a repeatable workflow for a known account and Workspace plan.
  - A fallback can preserve the same evidence, review and output requirements.
- **Avoid when**
  - A feature is assumed available because it appears in a demonstration or help page.
  - The workaround would copy restricted information into an unapproved account or tool.

## Quality Lens — Setting Up and Accessing Gemini Across Workspace Apps

### FAILURE SIGNAL

- The lab stops at a missing sparkle icon or unavailable control.

### REPAIR MOVE

- Name the job, use the documented fallback and capture the same final artifact.

### QUALITY EVIDENCE

- The access matrix shows account, app, feature, status and safe fallback.

# Effective Prompting for Everyday Work Tasks

## WHAT

- An effective prompt communicates four core elements: Persona, Task, Context and Format. The course adds a Source boundary and Review criteria so the response is useful for work: who Gemini should help, what to do, which facts to use, what the result should look like, and how a person will check it.

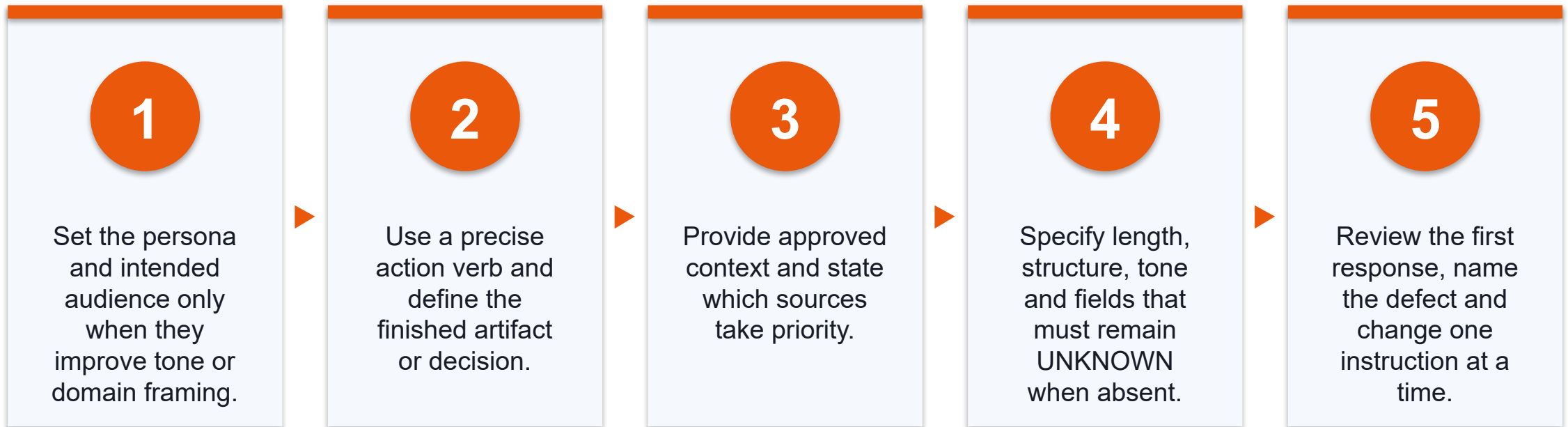
## WHY IT MATTERS

- Vague prompts hide assumptions and make quality subjective. A structured prompt creates an observable contract, supports iteration and lets another person reproduce the request rather than guessing what the author intended.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Effective Prompting for Everyday Work Tasks



## In Practice — Effective Prompting for Everyday Work Tasks

### WORKED EXAMPLE

- **Persona:** workplace coordinator writing for external partners.
- **Task and context:** draft a concise invitation using only the Aster & Finch project brief.
- **Format and review:** subject plus 150-word email; flag unsupported details and list a final fact check.

### DECISION GUIDE

- **Use when**
  - The task, audience, sources and desired structure can be stated clearly.
  - Several iterations can be compared against the same quality criteria.
- **Avoid when**
  - The prompt asks Gemini to invent missing evidence or imitate a real person's private style.
  - The user keeps adding conflicting instructions without resolving priorities.

## Quality Lens — Effective Prompting for Everyday Work Tasks

### FAILURE SIGNAL

- The output is generic, long or filled with invented specifics.

### REPAIR MOVE

- Add the missing P-T-C-F element, a source boundary and observable constraints.

### QUALITY EVIDENCE

- The final response satisfies a checklist another learner can apply consistently.

# Responsible, Secure and Private Use of AI at Work

## WHAT

- Responsible use combines data classification, least-privilege access, appropriate sharing, source verification, disclosure where needed and a human decision owner. Workspace protections matter, but they do not replace an organisation's policies or the user's duty to choose suitable inputs and recipients.

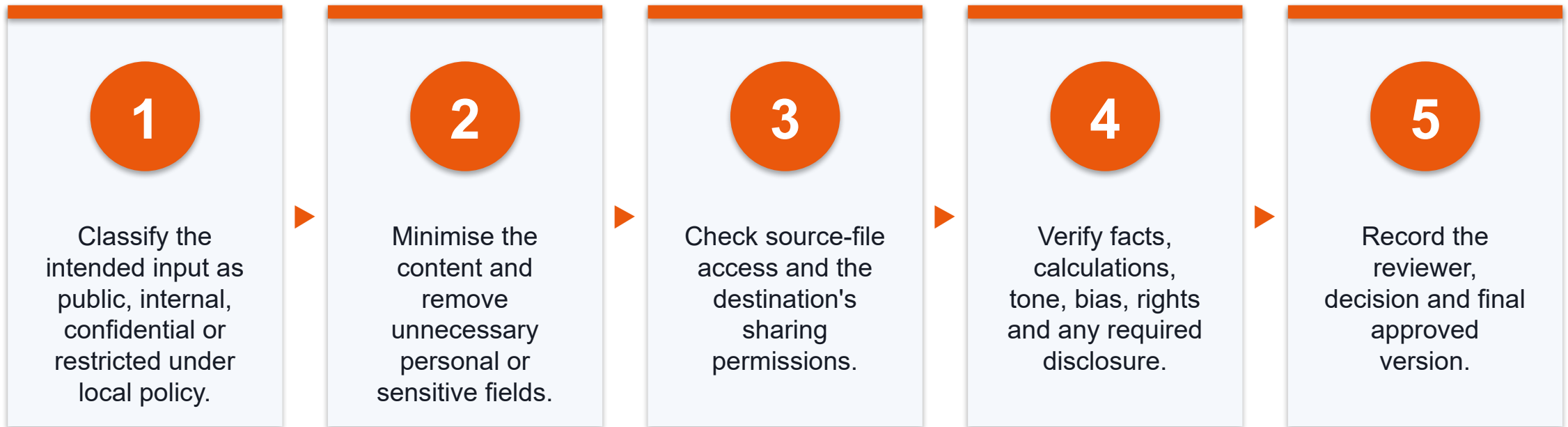
## WHY IT MATTERS

- Gemini can accelerate a mistake as easily as a good workflow. A privacy-safe prompt can still produce an inaccurate claim, and an accurate draft can still be overshared; controls must therefore cover the full path from source selection to final destination.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Responsible, Secure and Private Use of AI at Work





## In Practice — Responsible, Secure and Private Use of AI at Work

### WORKED EXAMPLE

- The synthetic registration sheet contains names and contact fields that the analysis does not need.
- The learner works from a minimised training copy, reports only grouped counts and shares results with named collaborators.
- A control log records sources, checks, approver and destination.

### DECISION GUIDE

- **Use when**
  - The organisation permits the data and task, and a defined reviewer can inspect the result.
  - The output remains inside an approved access and sharing boundary.
- **Avoid when**
  - The prompt includes passwords, secrets, protected personal records or information outside the user's authority.
  - The workflow hides uncertainty or makes a high-impact decision without qualified human review.

## Quality Lens — Responsible, Secure and Private Use of AI at Work

### FAILURE SIGNAL

- The draft is accurate but visible to people who do not need it.

### REPAIR MOVE

- Apply least privilege, remove unnecessary data and check the destination before sharing.

### QUALITY EVIDENCE

- The control log proves input authority, source checks, reviewer and recipient scope.

## Hands-On Labs — Getting Started with Gemini in Google Workspace

### Lab 1

- Confirm Access and Build a Safe Prompt Charter

### Lab 2

- Produce and Verify a Grounded Workspace Briefing

# Confirm Access and Build a Safe Prompt Charter

## LAB 1

You will create the C155 project folder, check Gemini and fallback availability across Workspace, classify the supplied synthetic sources and write a reusable P-T-C-F prompt charter. The charter keeps sources, uncertainty, sharing and human approval visible.

### You'll build

**A C155-Aster-Finch Drive folder with four state subfolders, an 01-access-matrix sheet and a 01-prompt-charter document containing approved sources, prohibited inputs, a reusable prompt pattern and the G-E-A-R review gate.**

1 Check access

2 Classify inputs

3 Write P-T-C-F

4 Set review gates

**Tools:** Google Drive - Docs - Sheets - Gemini features where available - supplied synthetic source pack

## Confirm Access and Build a Safe Prompt Charter

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### Test it

Open the Drive folder and both artifacts. There must be four correctly named state folders; 01-access-matrix must have seven app rows and four data-boundary rows; 01-prompt-charter must contain all eleven headings, one initial and one generated or manually refined second response, UNKNOWN for unsupported details, four G-E-A-R checks and a visible WORKING decision.

# Produce and Verify a Grounded Workspace Briefing

## LAB 2

You will apply the Lab 1 charter to create a structured partner-briefing candidate. You will then build a claim ledger, test the response against the sources and move only the reviewed version into the approved state.

### You'll build

**A 02-grounded-briefing Google Doc with seven required sections and a 02-claim-review-ledger Google Sheet that traces every material statement to a source, marks unsupported details and records the final human decision.**

1 Scope sources

2 Generate candidate

3 Trace claims

4 Approve or hold

**Tools:** Google Docs - Google Sheets - Gemini in Docs or supplied fallback - Drive

## Produce and Verify a Grounded Workspace Briefing

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### Test it

The briefing must contain all seven headings, 450-600 words, the two exact source paths, an open-question table and a four-row G-E-A-R table. The ledger must cover at least nine material claims and contain no VERIFIED row without a source excerpt. The Drive state must match the recorded APPROVED FOR COURSE WORKFLOW or HOLD decision.

## Recap — Getting Started with Gemini in Google Workspace

1

You can now: LO1: explain Gemini access, prompt structure and responsible controls across Google Workspace

2

You can now: LO1: use an effective prompt and a human review gate to create a source-grounded workplace briefing



TOPIC 02

# Gemini in Docs, Slides and Gmail

Drafting, rewriting and summarising - Slides and images - Gmail threads and replies - Reviewing generated content

## Key Concepts — Gemini in Docs, Slides and Gmail

1

### Source-led draft

A first version whose facts and constraints come from named approved material.

2

### Message architecture

Audience need, core message, supporting evidence and next action shape documents, slides and email.

3

### Visual brief

Purpose, subject, composition, aspect ratio, style and exclusions guide image generation.

4

### Review ledger

Claim, source, status, edit and approver make human refinement visible.

# Drafting, Rewriting and Summarising in Docs

## WHAT

- Gemini in Docs can create a starting draft, transform selected text and summarise information from a document or referenced Workspace sources. Drafting creates new structure, rewriting changes an existing passage, and summarising compresses meaning; each operation requires a different instruction and review test.

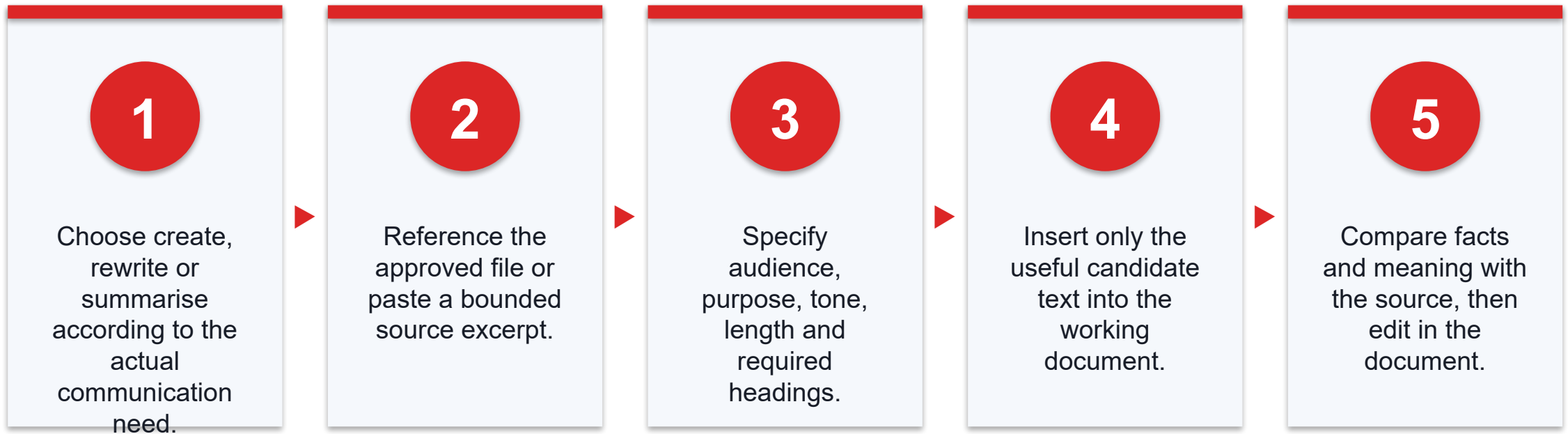
## WHY IT MATTERS

- Treating the three operations as interchangeable produces vague or distorted content. Naming the operation, audience and source boundary preserves intent and makes it easier to compare the generated passage with the evidence.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Drafting, Rewriting and Summarising in Docs



## In Practice — Drafting, Rewriting and Summarising in Docs

### WORKED EXAMPLE

- The coordinator asks for a one-page partner briefing using the project brief and facts file.
- Gemini produces Objectives, Audience, Agenda, Logistics and Open Questions.
- The coordinator rejects an invented capacity figure and keeps the verified structure.

### DECISION GUIDE

- **Use when**

- The source is known and the output is a draft that will be edited.
- A rewrite can be judged for meaning, tone and audience accessibility.

- **Avoid when**

- A summary would remove contractual nuance or a required qualification.
- The prompt gives only a title and expects correct organisation-specific facts.

## Quality Lens — Drafting, Rewriting and Summarising in Docs

### FAILURE SIGNAL

- A polished paragraph changes or omits a material source condition.

### REPAIR MOVE

- Quote the controlling source, require open questions and compare sentence by sentence.

### QUALITY EVIDENCE

- The document's claim ledger shows the source and final edit for every material fact.

# Generating Slides, Layouts and Images in Slides

## WHAT

- A useful presentation turns a decision or story into a sequence of claims, evidence and actions. Gemini can help create slide candidates, rewrite text and generate images, but the presenter still controls narrative, factual support, accessibility, brand fit and whether a generated visual represents reality appropriately.

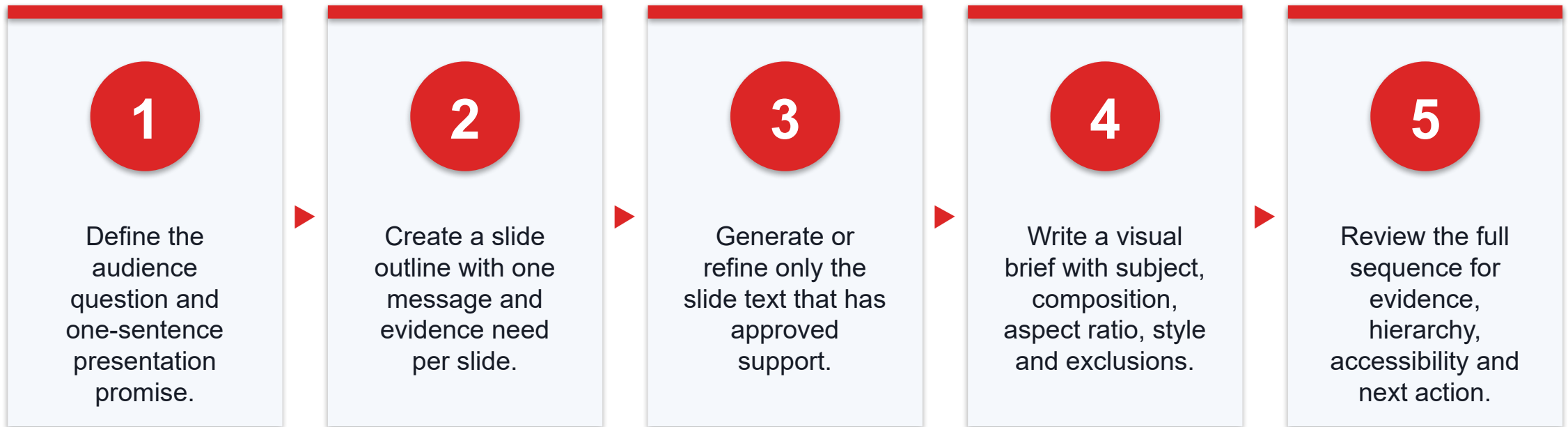
## WHY IT MATTERS

- A collection of attractive slides is not automatically a coherent presentation. Starting from audience and message architecture prevents decorative output from replacing the business purpose, while a visual brief reduces generic or misleading imagery.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Generating Slides, Layouts and Images in Slides





## In Practice — Generating Slides, Layouts and Images in Slides

### WORKED EXAMPLE

- The partner briefing uses six slides: purpose, audience need, agenda, participation data, action plan and next steps.
- A wide abstract collaboration image supports the title without depicting a real attendee or unsupported venue.
- All figures point to the registration sheet and all open details remain visibly pending.

### DECISION GUIDE

- **Use when**
  - The deck has a defined audience, message and evidence source.
  - Generated imagery is illustrative, appropriate and reviewed before use.
- **Avoid when**
  - The image could be mistaken for documentary proof of a real person, place or event.
  - Slide generation is used to hide missing analysis or compress unreadable amounts of text.

## Quality Lens — Generating Slides, Layouts and Images in Slides

### FAILURE SIGNAL

- Each slide looks acceptable alone but the sequence has no decision path.

### REPAIR MOVE

- Restate the audience question and give every slide one message and one role.

### QUALITY EVIDENCE

- A slide map links message, evidence, visual purpose and audience action.

# Writing, Replying and Summarising Emails in Gmail

## WHAT

- Gemini in Gmail can summarise a thread, draft a new message, suggest a reply and retrieve relevant information from permitted Workspace sources. The output must still respect thread history, recipient scope, commitments, tone and the difference between a suggested action and an approved promise.

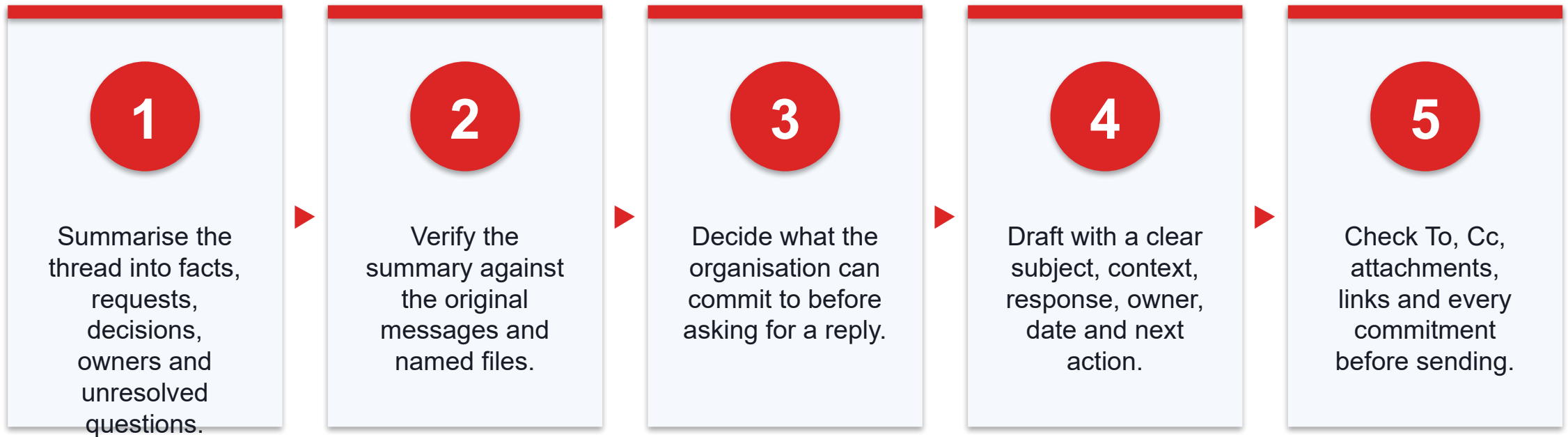
## WHY IT MATTERS

- Email is an external action surface: a small invented date, price or commitment can create real consequences. A safe workflow separates understanding the thread, deciding the response and drafting the message, then requires a final recipient and attachment check.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Writing, Replying and Summarising Emails in Gmail



## In Practice — Writing, Replying and Summarising Emails in Gmail

### WORKED EXAMPLE

- A synthetic partner thread asks about capacity, accessibility and the final agenda.
- Gemini separates confirmed facts from open questions and drafts a holding reply.
- The coordinator removes an unapproved promise and saves the message as a draft for review.

### DECISION GUIDE

- **Use when**

- The thread is bounded and the sender has authority to prepare a response.
- The final message can remain a draft until facts and recipients are checked.

- **Avoid when**

- The response would commit money, legal terms or sensitive information without an owner.
- The thread includes recipients or content the user is not authorised to expose to another tool or group.

## Quality Lens — Writing, Replying and Summarising Emails in Gmail

### FAILURE SIGNAL

- The draft promises a date or deliverable not agreed in the thread.

### REPAIR MOVE

- Convert unverified commitments to questions and name the decision owner.

### QUALITY EVIDENCE

- The saved draft matches the verified thread summary and has a completed send checklist.

# Reviewing and Refining AI-Generated Content

## WHAT

- Review is a structured comparison between the output, the source, the intended audience and the use context. The G-E-A-R loop used in this course checks Grounding, Evidence, Audience and Risk before a human approves, edits or rejects the candidate.

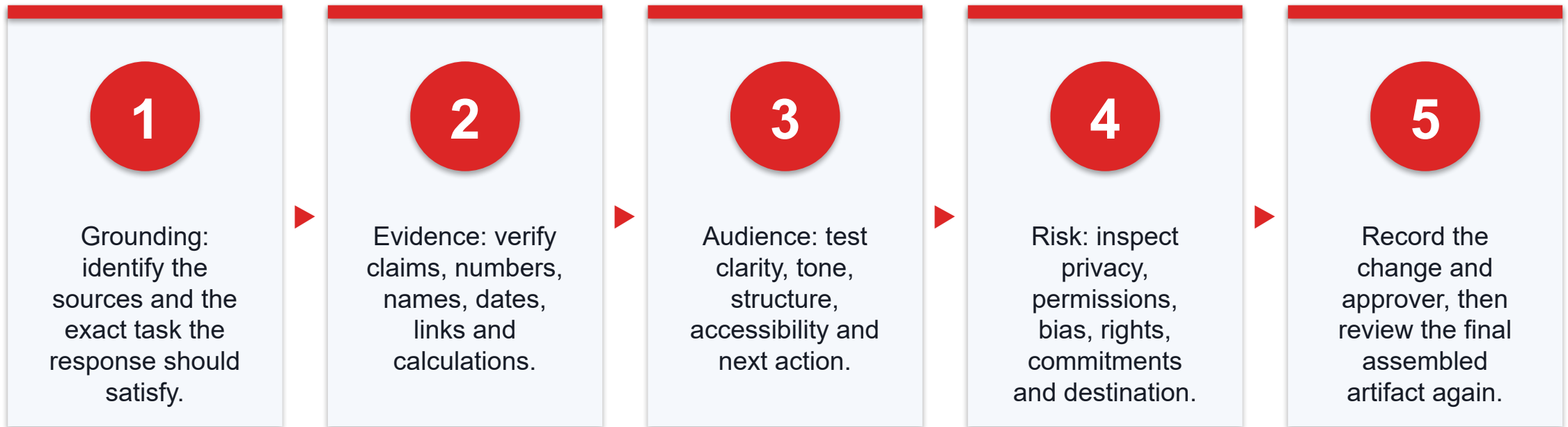
## WHY IT MATTERS

- Fluency makes generated content feel finished earlier than it is. An explicit loop catches factual drift, missing qualifications, inappropriate tone, inaccessible design, privacy exposure and unsupported actions across Docs, Slides and Gmail.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Reviewing and Refining AI-Generated Content





## In Practice — Reviewing and Refining AI-Generated Content

### WORKED EXAMPLE

- The deck states that 48 partners confirmed attendance, while the sheet contains 46 valid confirmations.
- The reviewer records the mismatch, corrects the slide and adds the sheet range to the source note.
- A second pass confirms the email and document use the same verified count.

### DECISION GUIDE

- **Use when**
  - The output will inform another person or become an organisation record.
  - A reviewer can access both source and final artifact.
- **Avoid when**
  - The review is reduced to spelling and visual polish only.
  - The final combined document is never checked after individually approved parts are merged.

## Quality Lens — Reviewing and Refining AI-Generated Content

### FAILURE SIGNAL

- Different Workspace artifacts contain different versions of the same fact.

### REPAIR MOVE

- Choose one controlling source, correct every consumer and re-run the final review.

### QUALITY EVIDENCE

- A cross-artifact ledger shows one verified value and every place it appears.

## Hands-On Labs — Gemini in Docs, Slides and Gmail

### Lab 3

- Draft the Partner Communication Pack in Docs and Gmail

### Lab 4

- Build the Partner Briefing Deck in Slides

## Draft the Partner Communication Pack in Docs and Gmail

### LAB 3

You will distinguish thread facts from requests and commitments, use Gemini to draft a partner invitation from approved sources, refine it in Docs and prepare a Gmail draft. The message remains unsent so recipient, attachment and commitment checks stay visible.

#### You'll build

**A 03-partner-communication-pack Google Doc containing a verified thread summary, partner invitation and internal hand-off note, plus one reviewed Gmail draft to the learner's own address with a completed send checklist.**

1 Summarise thread

2 Draft in Docs

3 Refine for audience

4 Save reviewed Gmail draft

**Tools:** Google Docs - Gmail - Gemini in Gmail or Docs where available - Drive

## Draft the Partner Communication Pack in Docs and Gmail

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### Test it

The communication pack must include all five headings, a thread summary that separates proposals from confirmed facts, a 180-220 word invitation, a five-field cross-artifact ledger, an internal hand-off with Owner/Action/Due/Source/Risk and a ten-row send checklist. Gmail must contain exactly one new unsent draft addressed only to the learner's own training account.

# Build the Partner Briefing Deck in Slides

## LAB 4

You will design a slide sequence before generating content, build six low-density slides from approved facts and create one illustrative visual with Gemini or a native-shape fallback. A slide map and G-E-A-R ledger keep the story, evidence and audience action aligned.

### You'll build

**A Google Slides file named 04-partner-briefing-deck with six slides, speaker notes and one reviewed visual, plus a slide-map table in 03-partner-communication-pack linking each slide to its message, source, visual purpose and audience action.**

1 Map the story

2 Create slide candidates

3 Build visual

4 Run sequence review

**Tools:** Google Slides - Gemini in Slides where available - Google Docs - Drive

# Build the Partner Briefing Deck in Slides

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## Test it

The Slides file must contain exactly six slides in the mapped order; slides 2-6 must each have a speaker-note source; no slide may exceed one headline and three bullets; one reviewed visual must have alt text; the preparation and workflow visuals must use the stated sequence; slide 3 must include the labelled DATA PLACEHOLDER; and all seven final review rows must be recorded under Deck review in 03-partner-communication-pack and show PASS.

# Recap — Gemini in Docs, Slides and Gmail

1 You can now: LO2: draft, summarise and refine source-led workplace content in Docs and Gmail

2 You can now: LO2: create and refine a coherent source-backed presentation and appropriate generated visual in Google Slides



## TOPIC 03

# Gemini in Sheets and Data Analysis

Formulas and tables - Analysis and summaries - Charts and visualisations - Cleaning and organising data

## Key Concepts — Gemini in Sheets and Data Analysis

1

### Data grain

What one row represents; every formula, summary and chart must respect that level.

2

### Reproducible transformation

A visible formula, rule or step that another person can rerun and inspect.

3

### Analysis question

A precise metric, population, dimension and time scope that makes a result interpretable.

4

### Chart contract

Title, axes, units, series and source must faithfully encode the stated comparison.

# Generating Formulas, Tables and Insights in Sheets

## WHAT

- Gemini in Sheets can help create tables, formulas and actions such as filters, formatting, pivots and dropdowns. The result is useful only when the data grain, field meaning, formula references and expected output are explicit and the inserted change is tested on known rows.

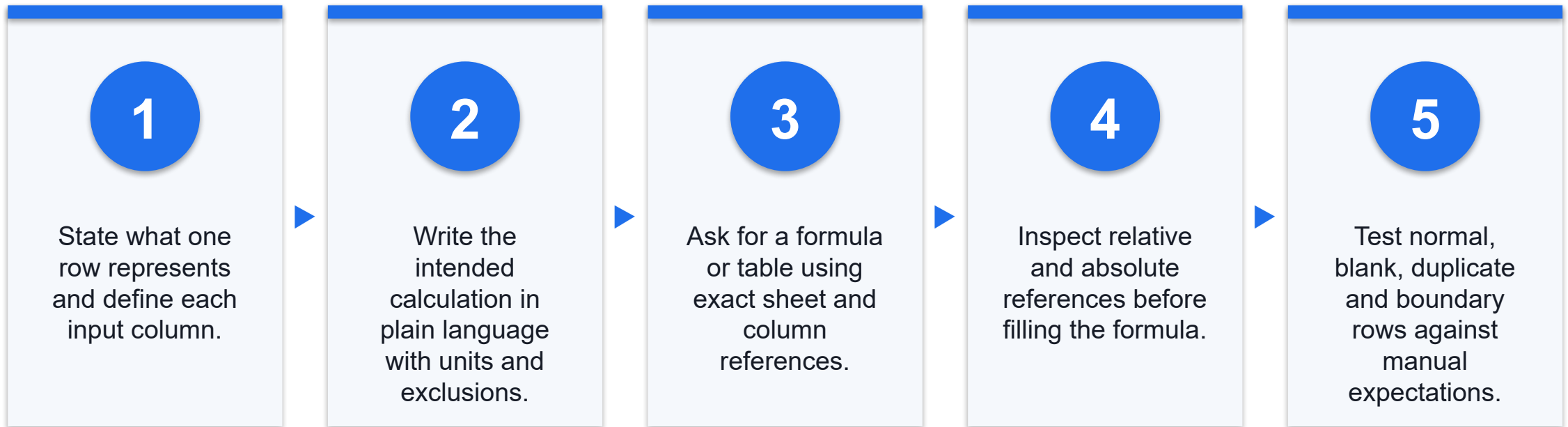
## WHY IT MATTERS

- A syntactically valid formula can still answer the wrong question. Starting with a plain-language calculation and sample expected result makes it possible to review a suggestion instead of accepting it because the cell displays a number.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Generating Formulas, Tables and Insights in Sheets



## In Practice — Generating Formulas, Tables and Insights in Sheets

### WORKED EXAMPLE

- Each row is one synthetic partner registration; Seats is the requested quantity.
- A clean formula standardises status and flags Confirmed rows with missing dietary information.
- The learner tests one confirmed, one waitlisted and one blank row before filling down.

### DECISION GUIDE

- **Use when**
  - The data fields and expected result can be described precisely.
  - The proposed formula remains visible and can be tested on sample rows.
- **Avoid when**
  - The sheet mixes several row types or merged headers without a defined grain.
  - A generated formula is filled across the dataset before its references are reviewed.

## Quality Lens — Generating Formulas, Tables and Insights in Sheets

### FAILURE SIGNAL

- The formula returns values but changes meaning when copied down.

### REPAIR MOVE

- Review row grain and lock only the references that should remain fixed.

### QUALITY EVIDENCE

- A test table shows input, expected result, actual result and status.

# Analysing and Summarising Data with Gemini

## WHAT

- Data analysis turns a stated question into defined metrics, comparisons and evidence. Gemini can propose pivots, calculations and narrative insights, but the analyst must define the population, denominator, filters, time scope and uncertainty before interpreting a pattern.

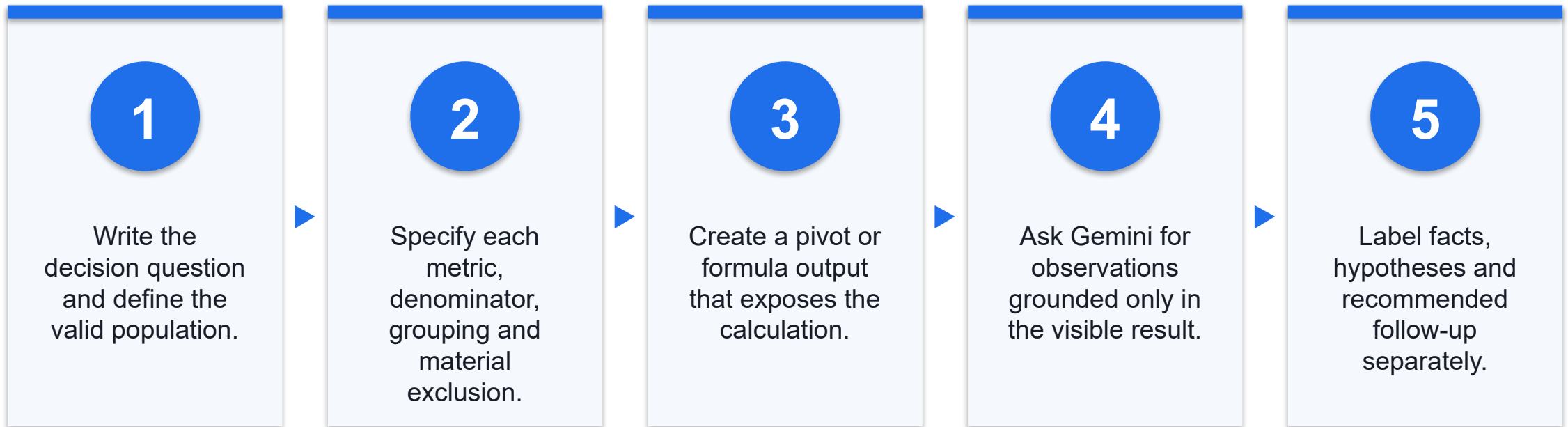
## WHY IT MATTERS

- Narrative summaries often sound causal even when the sheet only shows association. Keeping metric definitions and calculation evidence beside the insight prevents plausible wording from outrunning the data.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Analysing and Summarising Data with Gemini





## In Practice — Analysing and Summarising Data with Gemini

### WORKED EXAMPLE

- Question: which partner segment needs the most follow-up before the briefing?
- Metrics: valid registrations, confirmation rate and missing-information count by segment.
- The summary states the observed gap and proposes a follow-up check without claiming why it occurred.

### DECISION GUIDE

- **Use when**
  - The dataset is clean enough for the defined question and calculations are visible.
  - The narrative distinguishes observed evidence from a possible explanation.
- **Avoid when**
  - The prompt asks for insights before defining valid rows or metrics.
  - A small synthetic sample is used to claim a universal business trend.

## Quality Lens — Analysing and Summarising Data with Gemini

### FAILURE SIGNAL

- The insight has no metric definition or supporting range.

### REPAIR MOVE

- Add population, formula, grouping, time scope and source range.

### QUALITY EVIDENCE

- Another learner reproduces the value from the same clean table.

# Creating Charts and Visualisations

## WHAT

- A chart encodes a comparison through position, length, colour or shape. Gemini can propose and create visualisations in Sheets, but the user must choose a chart type that matches the question, display units and categories clearly, and verify every plotted value against the source table.

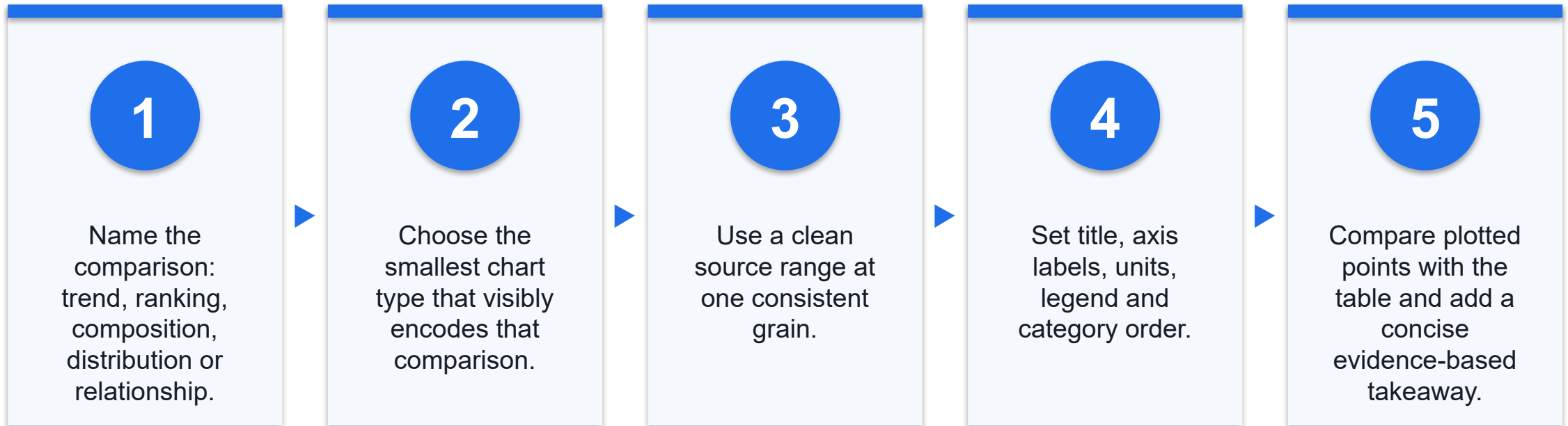
## WHY IT MATTERS

- Visuals compress data and therefore amplify both clarity and error. A neutral descriptive title, readable labels and an honest axis help viewers understand what is plotted without implying a conclusion the data cannot support.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Creating Charts and Visualisations



## In Practice — Creating Charts and Visualisations

### WORKED EXAMPLE

- A horizontal bar chart ranks valid registrations by partner segment.
- The title states 'Valid registrations by partner segment'; the x-axis shows registrations.
- The learner checks each bar against the pivot and avoids a truncated scale that exaggerates differences.

### DECISION GUIDE

- **Use when**
  - A visual makes three or more values or a pattern easier to compare.
  - The source table and metric definition can remain available to the reader.
- **Avoid when**
  - A single number would be clearer as text or a compact table.
  - A decorative 3D or dual-axis treatment could distort the comparison.

## Quality Lens — Creating Charts and Visualisations

### FAILURE SIGNAL

- The title claims a cause or trend that the chart does not encode.

### REPAIR MOVE

- Use a neutral title and put the bounded observation in a separate note.

### QUALITY EVIDENCE

- Title, axes, marks and source table all describe the same metric and grain.

# Cleaning and Organising Data with AI

## WHAT

- Data cleaning makes values consistent without hiding the original evidence. Typical operations include trimming spaces, standardising categories, parsing dates, handling blanks, identifying duplicates and validating allowed values; every change needs a rule and a retained raw source.

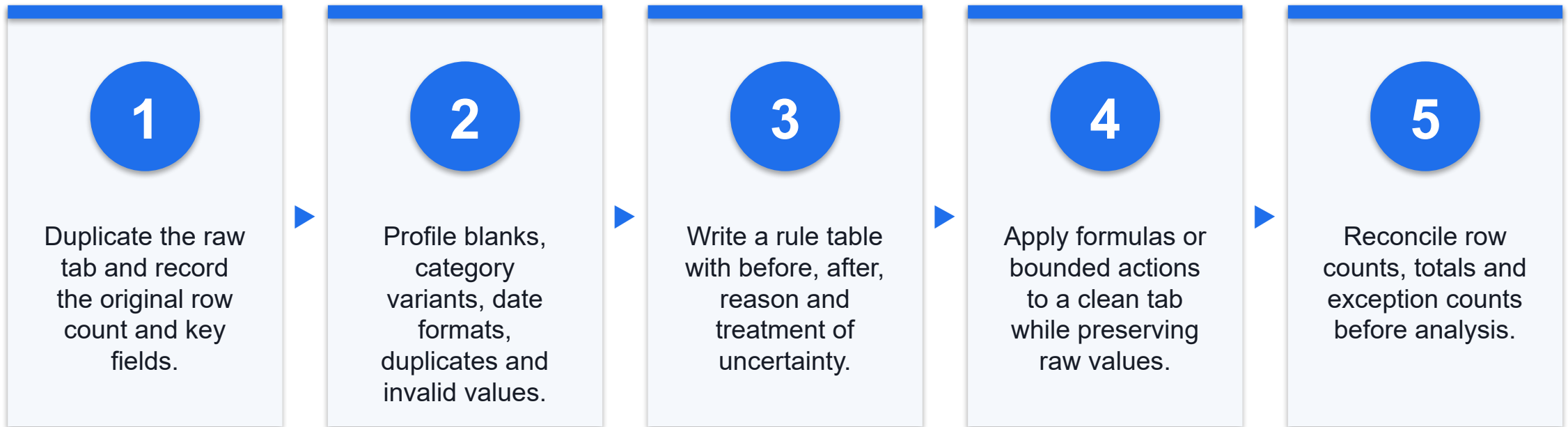
## WHY IT MATTERS

- Analysis quality is limited by the inputs. Asking Gemini to 'clean this' without rules can silently merge distinct values or fill unknowns, so a safe workflow profiles first, proposes rules, transforms a copy and reconciles row counts.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Cleaning and Organising Data with AI





## In Practice — Cleaning and Organising Data with AI

### WORKED EXAMPLE

- The raw Segment field contains SME, sme and Small business.
- The rule maps approved variants to SME but leaves an unfamiliar value as REVIEW.
- A reconciliation confirms that no valid registration disappeared during deduplication.

### DECISION GUIDE

- **Use when**
  - The organisation can define acceptable values and retain the raw source.
  - Exceptions can be reviewed instead of guessed.
- **Avoid when**
  - Missing values are silently invented or duplicate rules are not documented.
  - The only copy of the source is overwritten before totals are reconciled.

## Quality Lens — Cleaning and Organising Data with AI

### FAILURE SIGNAL

- The clean table has fewer rows and no explanation for the difference.

### REPAIR MOVE

- Restore the raw copy, log every excluded or merged row and reconcile totals.

### QUALITY EVIDENCE

- The rule table, exception list and reconciliation explain every change.

## Hands-On Labs — Gemini in Sheets and Data Analysis

### Lab 5

- Clean the Registration Tracker and Generate Formulas

### Lab 6

- Analyse Registrations and Build the Decision Dashboard

# Clean the Registration Tracker and Generate Formulas

## LAB 5

You will import a deliberately inconsistent registration CSV into Sheets, define the row grain and cleaning rules, ask Gemini for formula help and verify every formula before filling it down. A separate Valid tab and reconciliation make exclusions visible.

### You'll build

**A 05-registration-analysis Google Sheet with Raw, Clean, Valid, Rules and Formula-tests tabs. The workbook preserves 24 raw rows, produces 20 valid rows, flags exceptions and documents the exact formula and reason for every transformation.**

1 Profile raw data

2 Define rules

3 Apply formulas

4 Reconcile outputs

**Tools:** Google Sheets - Gemini in Sheets where available - Drive - registration-data.csv

## Clean the Registration Tracker and Generate Formulas

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### Test it

Raw must contain 24 data rows and remain unchanged; Clean must contain six new formula columns; Formula-tests must show six PASS rows; Valid must contain 20 data rows; and Rules must reconcile  $24 = 20 + 1 + 1 + 1 + 1$  with every excluded Registration\_ID named. A count of MISSING DIETARY in Valid must equal 3.

# Analyse Registrations and Build the Decision Dashboard

## LAB 6

You will define the valid population and metrics before asking Gemini for insights, create a pivot table and confirmation-rate calculations, then build a simple chart. The final narrative separates observations from hypotheses and updates the approved Slides deck.

### You'll build

**Analysis and Dashboard tabs inside 05-registration-analysis containing a verified pivot, metric definitions, confirmation rates, one horizontal bar chart and a three-part insight note, plus the linked chart inserted into slide 3 of 04-partner-briefing-deck.**

1 Define question

2 Build pivot

3 Verify metrics

4 Create chart and insight

**Tools:** Google Sheets - Gemini in Sheets where available - Google Slides - Drive

## Analyse Registrations and Build the Decision Dashboard

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### Test it

The pivot must reproduce 8/8/4 valid registrations and 5/5/2 confirmations by segment; the rates must be 62.5%, 62.5% and 50.0%; total missing information must be 3; the chart must have one series, a zero baseline and the neutral title; the insight must separate observation, hypothesis and follow-up; and slide 3 must contain the linked chart with the population and exclusion source note.

# Recap — Gemini in Sheets and Data Analysis

1

You can now: LO3: clean and organise spreadsheet data with documented rules, Gemini suggestions and reproducible formulas

2

You can now: LO3: define metrics, analyse clean data and create a truthful chart and bounded narrative in Sheets



## TOPIC 04

# AI-Powered Workflows and Automation in Workspace

Meet notes - Drive search and organisation - Connecting apps and simple automations - Practical workflow design

## Key Concepts — AI-Powered Workflows and Automation in Workspace

1

### Workflow state

Input, owner, action, output and approval status make work visible across apps.

2

### Permission-aware retrieval

Gemini can only be useful and safe when sources and recipients match authorised access.

3

### Bounded automation

A trigger performs a narrow action with explicit fields, logs and stop conditions.

4

### Observe before scaling

Run, inspect evidence, handle exceptions and expand only after a controlled test.

# AI-Assisted Meetings and Notes in Google Meet

## WHAT

- Eligible Google Workspace accounts can use Gemini-supported meeting features to capture notes, summarise discussion and identify next steps. The notes are a generated record that must be announced, shared deliberately and reviewed against the conversation before becoming an official source.

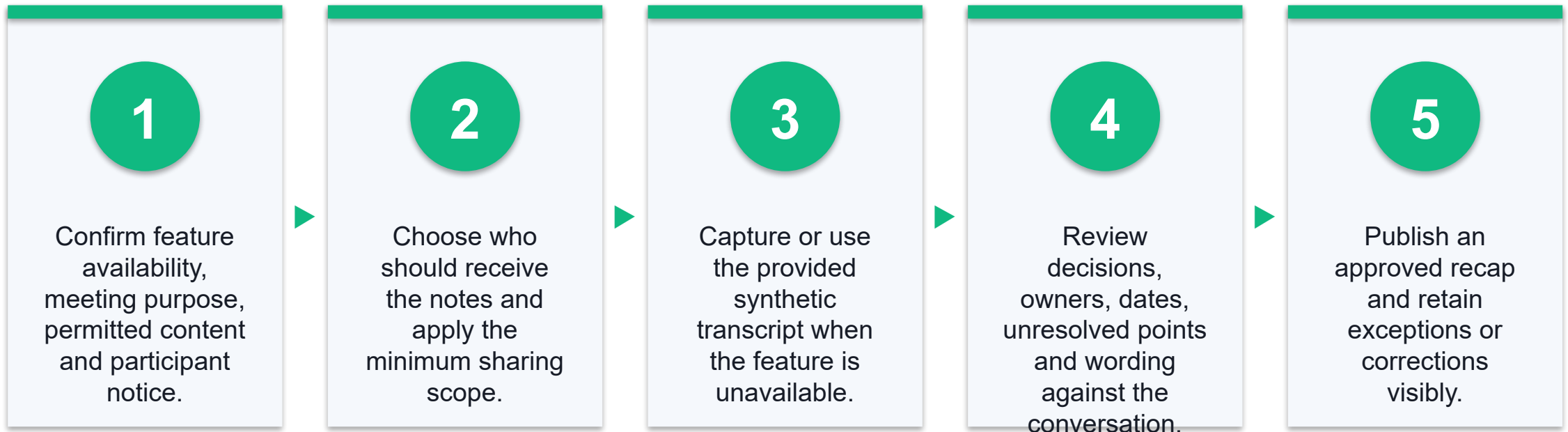
## WHY IT MATTERS

- Meeting summaries can save transcription and follow-up time, but they may omit context, misattribute an action or expose notes to the wrong invitees. Consent, host control, language support, sharing settings and post-meeting review are part of the workflow rather than optional administration.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — AI-Assisted Meetings and Notes in Google Meet



## In Practice — AI-Assisted Meetings and Notes in Google Meet

### WORKED EXAMPLE

- The partner-briefing rehearsal has a synthetic ten-minute transcript and three proposed actions.
- The coordinator corrects one owner, changes an uncertain date to pending and restricts the notes to the internal team.
- Only the approved action table is copied to the shared follow-up document.

### DECISION GUIDE

- **Use when**
  - Participants are informed and the organisation permits the meeting feature.
  - A host or owner will review the generated notes before wider sharing.
- **Avoid when**
  - The conversation contains restricted material outside the approved note-taking purpose.
  - Invite visibility is mistaken for permission to access the generated notes document.

## Quality Lens — AI-Assisted Meetings and Notes in Google Meet

### FAILURE SIGNAL

- A generated action item has the wrong owner or an invented deadline.

### REPAIR MOVE

- Check the relevant transcript segment and mark unresolved fields as pending.

### QUALITY EVIDENCE

- The review log records original text, correction, source and approver.

# Organising and Searching Drive with Gemini

## WHAT

- Gemini in Drive can summarise files or folders, answer questions across permitted sources and help locate or organise material. Retrieval is permission-aware but still depends on clear folder scope, current file versions, meaningful names and a user who verifies the cited sources.

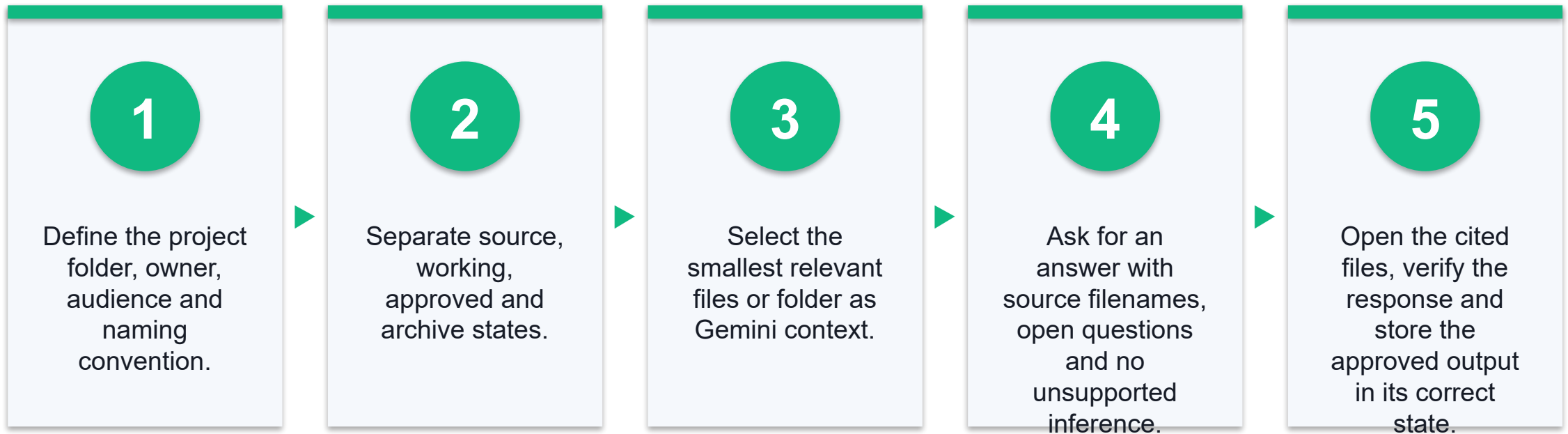
## WHY IT MATTERS

- Search becomes unreliable when a folder mixes drafts, duplicates and unclear ownership. A simple information architecture and source register let Gemini retrieve relevant evidence without treating an obsolete file as the controlling record.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Organising and Searching Drive with Gemini





## In Practice — Organising and Searching Drive with Gemini

### WORKED EXAMPLE

- The C155 project folder contains 01-Source, 02-Working, 03-Approved and 04-Archive.
- Gemini summarises only the approved brief and reviewed registration insight, citing both filenames.
- An older agenda remains in Archive and is not used as the final event source.

### DECISION GUIDE

- **Use when**
  - The folder has clear ownership, versions and access controls.
  - The response can show which files supplied the relevant information.
- **Avoid when**
  - A broad Drive query could mix personal, obsolete or unrelated content.
  - The user assumes retrieval proves the answer is complete or current.

## Quality Lens — Organising and Searching Drive with Gemini

### FAILURE SIGNAL

- The answer cites a draft that should no longer govern the project.

### REPAIR MOVE

- Clarify file states, narrow the source scope and identify the controlling record.

### QUALITY EVIDENCE

- Every answer links to an accessible current source in the project register.

# Connecting Apps and Building Simple Automations

## WHAT

- Google Apps Script is a cloud-based JavaScript platform for extending and automating Workspace. A simple automation reads defined fields, applies a narrow rule and creates an observable action such as a Gmail draft or status update; it should include a dry-run mode, log and explicit authorisation.

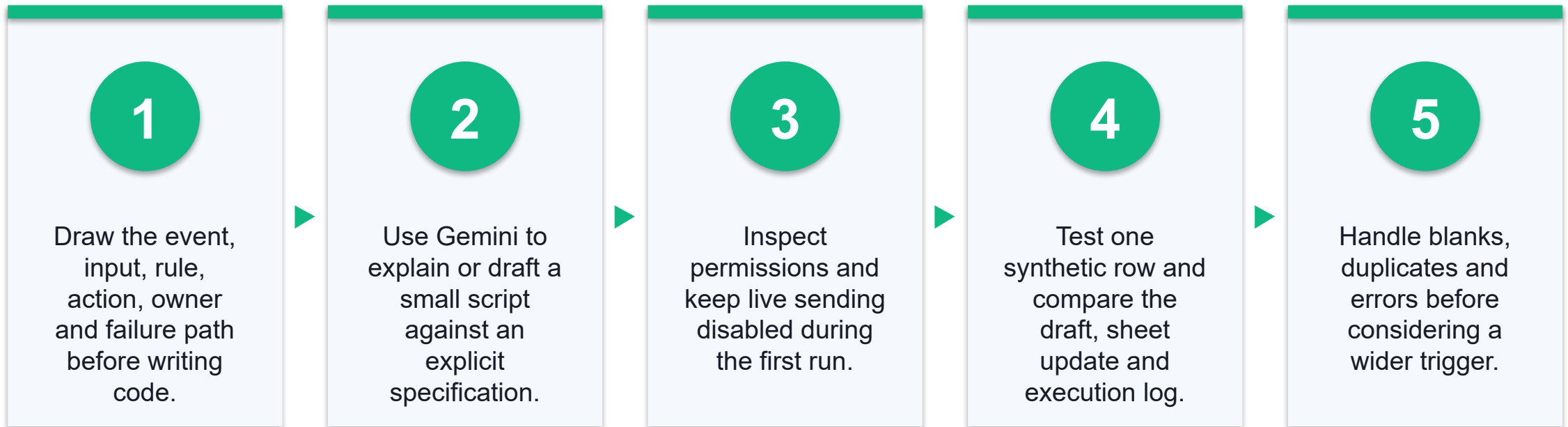
## WHY IT MATTERS

- Automation removes repeated clicks but also repeats defects at speed. Restricting the trigger, recipient, data fields and action surface makes a beginner workflow easier to understand, test, stop and review before it affects other people.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Connecting Apps and Building Simple Automations



## In Practice — Connecting Apps and Building Simple Automations

### WORKED EXAMPLE

- A bound Sheets script reads one READY row and creates a Gmail draft to the learner's own address.
- Dry-run mode records the intended recipient and subject without creating the draft.
- A live test creates one draft, writes DRAFTED and stores a timestamp so the row is not processed twice.

### DECISION GUIDE

- **Use when**

- The process is repetitive, rule-based, low-risk and has an observable success condition.
- The user can review scopes, logs and every action before expanding the run.

- **Avoid when**

- The automation would bulk-send, delete, change access or process confidential data as a first test.
- The workflow lacks an owner, idempotency rule or way to stop and inspect failures.

## Quality Lens — Connecting Apps and Building Simple Automations

### FAILURE SIGNAL

- Running the script twice creates duplicate external actions.

### REPAIR MOVE

- Add a processed status or unique key and skip rows already handled.

### QUALITY EVIDENCE

- The test log shows one input row, one intended action and one final status.

# Designing Practical AI Productivity Workflows

## WHAT

- A practical workflow links a business trigger to approved sources, human and AI steps, outputs, controls and evidence. The C-O-N-T-R-O-L canvas used here captures Context, Owner, Needed inputs, Task sequence, Review, Output destination and Learning signal.

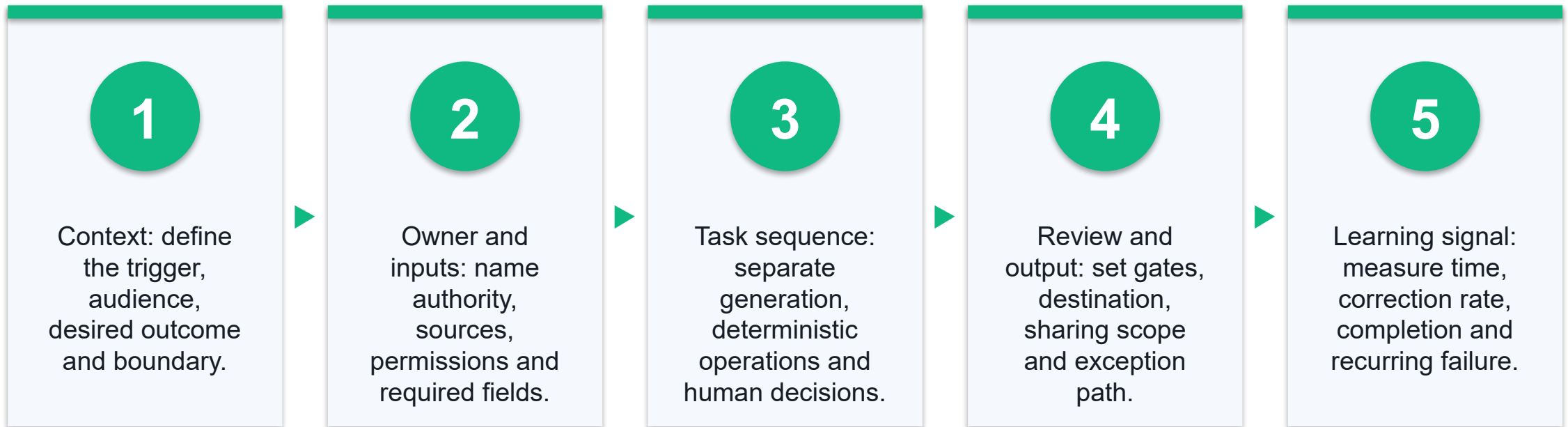
## WHY IT MATTERS

- Tool-first automation often optimises an unclear process. Designing the state changes and review gates first reveals where Gemini adds value, where a person must decide, and which evidence proves the workflow is safe and useful.

## HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

## How It Works — Designing Practical AI Productivity Workflows





## In Practice — Designing Practical AI Productivity Workflows

### WORKED EXAMPLE

- A READY registration row triggers a proposed follow-up, not an automatic external send.
- Gemini helps draft the message; Apps Script creates a Gmail draft; the coordinator reviews and sends manually.
- The log records draft creation, corrections and whether the workflow saved time without increasing errors.

### DECISION GUIDE

- **Use when**
  - The process has a stable trigger, bounded inputs and a human owner.
  - Success, correction and exception evidence can be collected.
- **Avoid when**
  - The process is still ambiguous or relies on unrecorded personal judgement.
  - Productivity is measured only by output volume rather than quality and risk.

## Quality Lens — Designing Practical AI Productivity Workflows

### FAILURE SIGNAL

- The workflow is fast but no one can explain who approved the result.

### REPAIR MOVE

- Add a named gate, status field and evidence artifact before the output leaves Draft.

### QUALITY EVIDENCE

- The canvas and run log link trigger, sources, actions, reviewer, output and learning.

## Hands-On Labs — AI-Powered Workflows and Automation in Workspace

### Lab 7

- Run the Meeting-to-Drive Follow-Up Workflow

### Lab 8

- Automate the Approved Follow-Up from Sheets

# Run the Meeting-to-Drive Follow-Up Workflow

## LAB 7

You will configure or simulate a Gemini-supported meeting-notes workflow, review a supplied transcript, correct owners and dates, and organise the project's approved Drive sources. Gemini in Drive or a manual fallback then produces a source-cited follow-up brief.

### You'll build

**A 07-meeting-notes-review Google Doc, a 07-source-register Google Sheet and a 07-approved-follow-up-brief Google Doc containing three verified actions, one open question, source citations, sharing controls and a correction log.**

1 Set note controls

2 Review transcript

3 Organise Drive

4 Synthesize approved follow-up

**Tools:** Google Meet - Gemini meeting features where available - Google Drive - Docs - Sheets

## Run the Meeting-to-Drive Follow-Up Workflow

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### Test it

The meeting-notes review must show control settings, three verified actions owned by Jamie, Marcus and Noor, one facilitator open question and a correction log. The source register must cover five current project artifacts with state, owner, range and access. The follow-up brief must cite exact filenames, retain 20 valid/4 excluded and 8/8/4 segment counts, use no Archive source and have a sharing setting that matches the recorded decision.

# Automate the Approved Follow-Up from Sheets

## LAB 8

You will use the approved follow-up evidence to specify a low-risk automation, ask Gemini to explain or draft the code and compare it with the supplied safe script. When Apps Script is available, the script first logs an intended action, then creates one Gmail draft to your own address and refuses to duplicate it. When access is blocked, you trace the same controls and clearly leave live evidence pending.

### You'll build

**An Automation tab and CONTROL canvas in 05-registration-analysis plus either (a) a bound Apps Script project using apps-script-draft.gs, one reviewed Gmail draft to the learner's own account and a run log, or (b) a four-case code trace with live evidence explicitly pending. Both paths cover dry-run, live-run, HOLD and duplicate-prevention**

1 Design CONTROL canvas

2 Prepare one row

3 Run or trace dry test

4 Verify draft or blocked-path  
evidence

**Tools:** Google Sheets - Apps Script - Gmail Drafts - Gemini in Workspace where available

## Automate the Approved Follow-Up from Sheets

### ✓ Test it

CONTROL must define all seven fields and the stated limits; Automation must contain one READY self-addressed row and one HOLD synthetic row. On the available path, dry run must create no draft, live run must create exactly one unsent C155 TEST Gmail draft and write DRAFTED, ID and timestamp, the repeat run must create no duplicate and HOLD must remain untouched. On the blocked path, CONTROL must trace dry READY self, live READY self, repeat DRAFTED and HOLD with the expected branch, Sheet writes and Gmail effect; live evidence fields must remain blank and the decision must be CODE WALK-THROUGH COMPLETE - LIVE TEST PENDING. The relevant run record and final human decision must be complete.

## Recap — AI-Powered Workflows and Automation in Workspace

1

You can now: LO4: review AI-assisted meeting notes, organise approved Drive sources and produce a permission-aware follow-up brief

2

You can now: LO4: design, implement and verify a bounded Apps Script automation that connects Sheets and Gmail





WRAP-UP

# Course Summary & Next Steps

# What You Achieved

1

## Prompt & Protect

Explain how Gemini works across Google Workspace, write effective prompts and apply responsible data, privacy and verification controls.

2

## Write & Present

Draft, rewrite, summarise and refine workplace content in Docs, Slides and Gmail while preserving source accuracy, audience fit and human approval.

3

## Analyse & Visualise

Clean, organise, analyse and visualise spreadsheet data with Gemini in Sheets, formulas and reproducible checks.

4

## Connect & Automate

Design an AI-assisted workflow across Meet, Drive, Sheets, Docs and Gmail, then implement and verify a bounded Workspace automation.

# A Practical 30-Day Adoption Plan

1

Week 1 - inventory one repetitive task, its approved sources, data class, owner and current failure points.

2

Week 2 - test a P-T-C-F prompt and G-E-A-R review on synthetic or minimised data.

3

Week 3 - pilot a draft-only cross-app workflow with logs, exceptions and a small authorised group.

4

Week 4 - compare time saved, corrections, completion and risk observations before approving the next scope.

# Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Keep your downloaded C155 course files and use the lab checkpoints to restart safely.

4

Explore the follow-on courses at [www.tertiarycourses.com.sg](http://www.tertiarycourses.com.sg).

PROMPT WITH PURPOSE - VERIFY WITH EVIDENCE - ACT WITH CONTROL

# Thank You!

You can now use Gemini across Google Workspace as a controlled collaborator: prompt with context, verify with evidence, share deliberately and automate in small observable steps.